Tarek **Eldeeb**

R&D Manager · FPGA Expert
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Scan to Update

"Servant leader, lecturer, creative geek with roots in electronics engineering, and an entrepreneurial mindset."

Education

Executive Masters of Business Administration — *Nile University*

Graduated with GPA 3.78 (48 credit hours) — *2018 - 2020*

- The EMBA program is designed in collaboration with IESE Business School, University of Navarra, Spain.

B.Sc. in Electronics and Communications Engineering — *Cairo University*

Graduated with grade Good — *2001 - 2006*

- Graduation project was graded Excellent. On the IEEE's annual EED event, the project was awarded as 'Best Project'.
- Passed several post-graduate classes including: Advanced Computer Networks, Computer Architecture, Computer Arithmetic, and Advanced Mathematics.

Skills

Management	Visionary Technical Leadership, Business development, SAFe Agile Project and Product Management
Electronics	VHDL/Verilog, AXI-based SoC, FPGA, Microblaze, MGC Questa, PCB
Programming	ANSI C, RTOS, Linux drivers, Java, Python, Matlab, Tensorflow, Bash, TCL, CI/CD, LaTeX, Git
Languages	Arabic, English

Experience

Chief Technical Officer — *Techno-Welle*

Multi-site Role — *Jul. 2024 - Current*

- Drive business growth wave.
- Building Technical Teams.
- Launched multiple products, with successfully closing sales deals:
  - Industrial End-of-line Tester: Negotiated and closed B2B sales contracts in annual recurring revenue (ARR).
  - Fleet Management System: Spearheaded the go-to-market strategy for Fleet Marschall, closing the first succesfull deal.
  - IIoT: Aligned cross-functional engineering and sales teams to accelerate product delivery.

Technical Product Manager — *Valeo*

Engineering Excellence — *Sep. 2023 - Jun 2024*

- SAFe® Product Manager for Veleo's private HIL cloud product; a group-wide standard product directly monitored by the CTO.
- Realize business needs into practical technical solutions. This covers DevOPs, Cloud Apps, IoT, and Cyber security technologies.
- Drive Product Increment Events and manage the project backlog.

Senior SW R&D Manager — *Valeo*

Comfort & Driving Assistance Business Group — *Oct. 2016 - Aug. 2023*

- Put the strategy for Valeo's SW validation test bench, securing funds, and aligning plans execution across different sites.
- Conduct a frequent analysis exercise to assess our rivals, enhance our quality and secure the supply chain.
- Lead and manage a test bench service-line serving 700 engineer worldwide.
- Parallel Agile Projects Management, do resource planning and enhance customer satisfaction.
- Successfully deliver HIL test benches to 50+ projects at 6 R&D sites and 3 production sites.

FPGA Expert — *Valeo*

A business group level secondary title — *Apr. 2019 - Jun. 2024*

- Architect, Lead engineer and product owner of Valeo's internal SW validation Hardware In then Loop 'HIL' Test bench.
- Support Valeo's first ASIC design project.
- Valeo University relations: FPGA Class Lecturer and owner at ITI, during 2019-2023.
- Provides consultations and training to several product lines and R&D sites.

Senior SW Team Lead

Valeo

Comfort & Driving Assistance Business Group — *July. 2013 - Sep. 2016*

- Implement HD video streams, high-performance ultrasonic sensors simulation, Ethernet, Flexray and CAN bus communication using FPGA IPs in VHDL/Verilog
- Design test boards; PCB schematics design and layout
- Develop and maintain an RT kernel with firmware and drivers, in addition to a Java-based Test-Automation Environment
- Enable Automotive SW Validation for automatic self-parking cars, Laser Radars, Video surround views and many advanced driver assistance systems

Cofounder and Digital Design Engineer

SilMinds

A 6-year startup journey — *May 2007 - Jun. 2013*

- Design and implement decimal floating-point arithmetic IP cores comforting the IEEE754-2008 standard.
- VHDL implementation, data path optimization, static timing analysis, FPGA tailoring, and writing automation scripts for EDA tools.
- Consultant at two VLSI companies (Si-Vision and VarkonSemi) to develop a configurable dynamic testing environment for their next generation chips using FPGAs.

Testing Lab Engineer

SysDSoft

WiFi Data Link Layer Controller — *Aug 2006 - May 2007*

- Executed test scripts to automate hardware-in-loop validation for our wireless 802.11 a/b/g/q chips.
- SysDSoft was acquired by Intel (2011), then sold to Apple (2019).

Publications

2022	**Research Paper**, "On-line Hybrid Perceptual and Cryptographic Image Hashing System"	*ISEEIE, Thai*
2018	**Research Paper**, "Dynamic Code Loading to a Bare-metal Embedded Target"	*ICSIE, Cairo*
2014	**Patent**, "Decimal Floating-Point Fused-Multiply-Add Unit"	*USPTO, US*
2014	**Patent**, "Decimal Floating-Point Square-Root Unit Using Newton-Raphson iterations"	*USPTO, US*
2014	**Patent**, "Rounding Unit for Decimal Floating-Point Division"	*USPTO, US*
2014	**Patent**, "Parallel Redundant Decimal Fused-Multiply-Add Circuit"	*USPTO, US*
2014	**Patent**, "Decimal Elementary Functions Computation"	*USPTO, US*
2012	**Patent**, "DPD/BCD To BID Converters"	*USPTO, US*
2012	**Patent**, "Decimal Floating-Point Processor"	*USPTO, US*
2011	**Book Chapter**, "Massively Parallelized DNA Motif Search on FPGA"	*IntechOpen, UK*
2010	**Research Paper**, "Decimal Floating Point for future processors "	*ICM, Cairo*
2010	**Research Paper**, "Algorithm and architecture for on-line decimal powering computation"	*ASILOMAR, CA, US*
2010	**Research Paper**, "A Decimal Floating-point Fused-Multiply-Add Unit"	*MWSCAS, WA, US*
2009	**Research Paper**, "Energy and Delay Improvement via Decimal Floating Point Units"	*Comp. Arith., OR, US*
2008	**Research Paper**, "A decimal fully parallel and pipelined floating point multiplier"	*ASILOMAR, CA, US*

According to Google Scholar Profile, there are 150 citation to my work.

Training

2024	**SAFe6 Practitioner**, Scaled Agile, Inc.	*Certificate*
2022	**Change Management**, Logic Training	*Certificate*
2021	**Embedded Linux**, Linux Hotel	*Certificate*
2020	**Managing Remotely**, Impact Learning and Development	*Certificate*
2019	**Certified KPI Professional**, KPI Institute	*Certificate*
2018	**Deep Learning Specialization**, DeepLearning.ai (coursera)	*Certificate*
2018	**Develop Collaborative Management - Hofstede Model Approach**, Hostede Insights	*Certificate*
2017	**Leadership Program**, Redrock International	*Certificate*
2017	**Linux Kernel Driver Programming with Embedded Devices**, Udemy	*Certificate*
2014	**Advanced Time Management**, Brilliance Business School	*Certificate*
2013	**Train the Trainer**, Brilliance Business School	*Certificate*
2010	**Agile Project Management**, The International Consortium for Agile	*Certificate*
2009	**Project Management Program**, RS Management Consulting House	—

All references are embedded in this PDF file, please click or above.